

Reference Variable ÷

In C/C++

`int x = 5;`

x
5

↓
ordinary variable

`int *p;` → pointer variable

It will hold the address of any variable

There is one more variable in C++.

`int &y = x;`

↳ Reference variable

For our understanding we can say that -

Making a reference variable means giving another name to variable x.

Important point ÷

Once a variable is declared as **reference variable** we cannot update / modify it. We can only control the value of the variable whose reference the **reference variable** is holding.

Ex - `int x = 5;`
`int &y = x;`
`y++;`
`cout << x;`

Output -

6



1. Reference variable must be initialized during declaration. It can be initialized with already declared variables only.
2. Reference variable can not be updated.

int x = 10;
int y = x;

Reference variable x is initialized with value 10.
Reference variable y is initialized with value of x (10).

Both x and y are reference variables.

x = 12; // Error
Reference variable x cannot be updated.

Reference variable x is initialized with value 10.
Reference variable y is initialized with value of x (10).
Reference variable x cannot be updated.

int x = 10;

Reference variable x is initialized with value 10.
Reference variable y is initialized with value of x (10).
Reference variable x cannot be updated.

x = 10;
y = x;
x = 20; // Error

int x = 10;